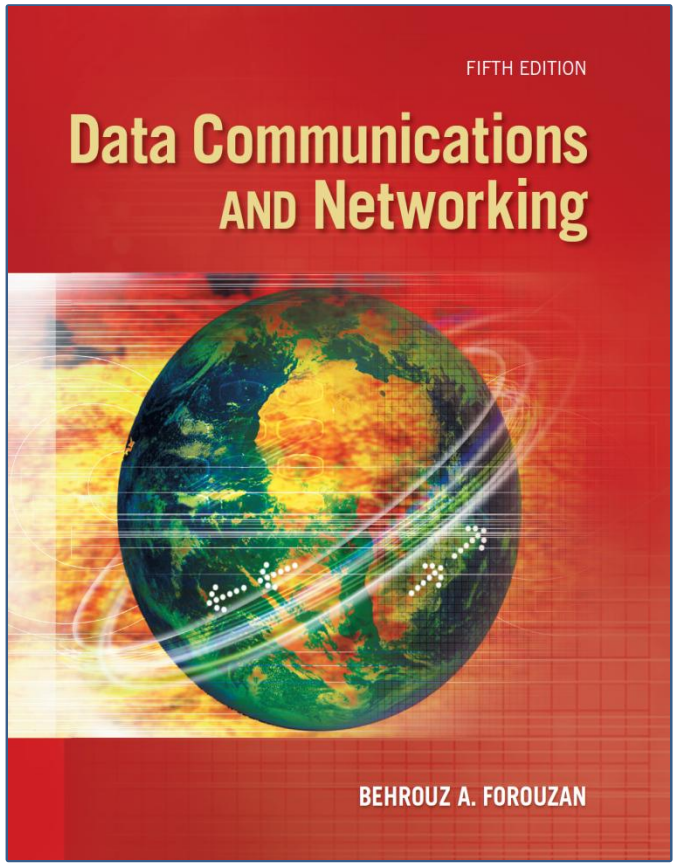



# Network Services

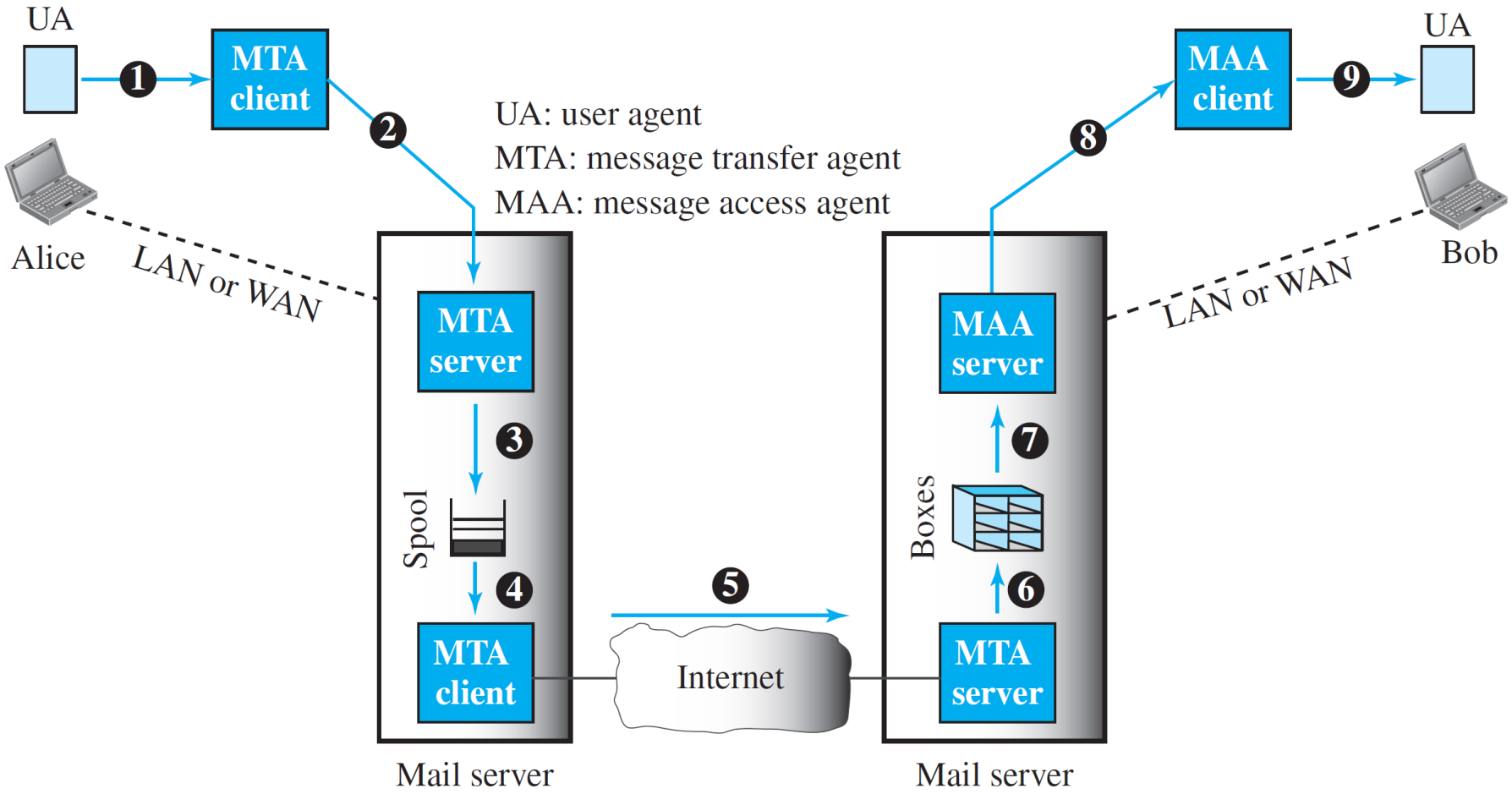
Lecturer: Dr. Khaled BinAbdl

# Electronic Mail System

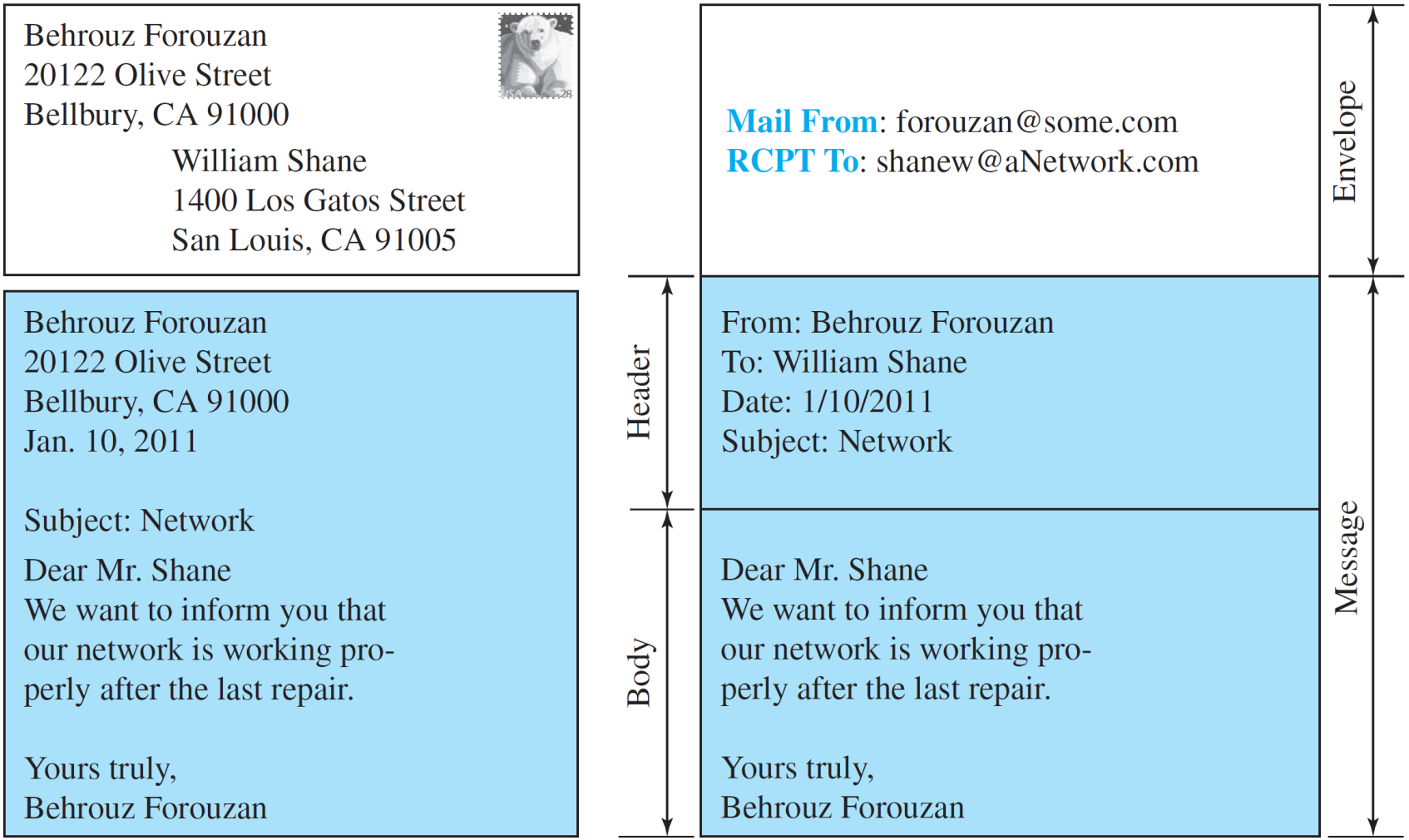
- **Electronic mail** (or e-mail) allows users to **exchange messages**.

- Email service **properties**:
  - Is considered as a one-way transaction.
  - The recipient does not wait until someone sends an e-mail to him.

# Email Architecture



# Format of an Email



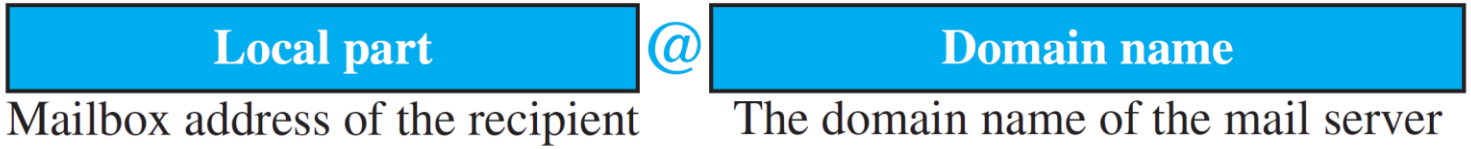

Postal mail

Electronic mail

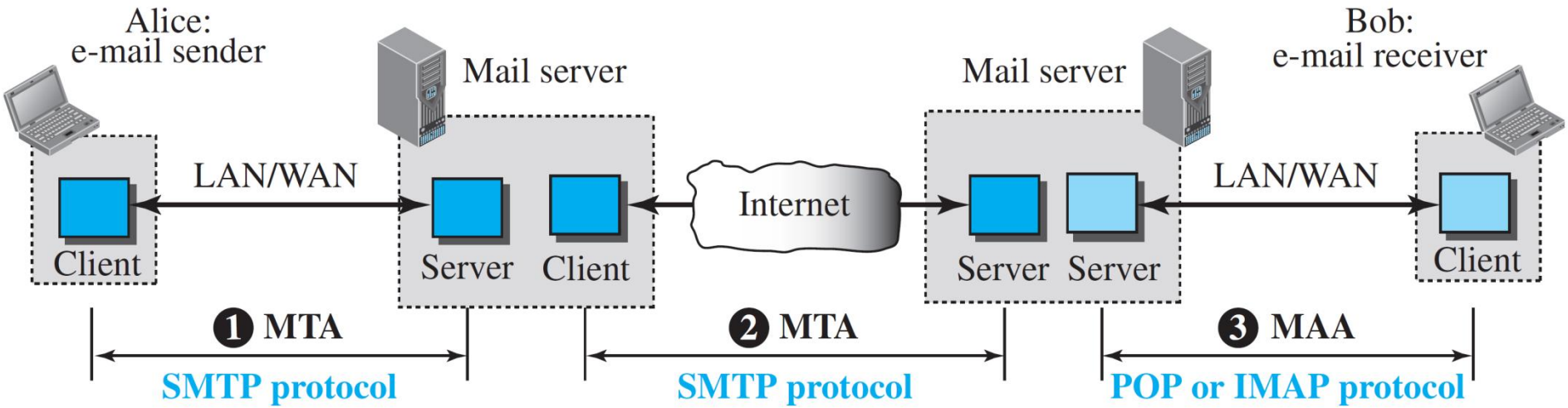
# Email Addressing

– In the **Internet**, email **address** consists of **two** parts **separated** by an @ sign :

- *local part*
- *domain name*



# Message Transfer Agent: SMTP



# Simple Mail Transfer Protocol (SMTP).

– SMTP is the **formal** protocol that **defines** the **MTA** client and server in the Internet.

– SMTP is used **two times**:
  - between the sender and the sender's mail server
  - between the two mail servers.

# Commands and Responses

- SMTP uses **commands** and **responses** to transfer messages **between** an MTA **client** and an MTA **server**.
  - The **command** is **from** an MTA client **to** an MTA server;
  - The **response** is **from** an MTA server **to** the MTA client.

- Each **command** or **reply** is **terminated** by a **two** character (**carriage return** and **line feed**).

# SMTP Commands

– **Commands** are **sent** from the **client** to the **server**.

**Keyword:** argument(s)

# SMTP Commands

| *Keyword* | *Argument(s)* | *Description* |
|---|---|---|
| HELO | Sender's host name | Identifies itself |
| MAIL FROM | Sender of the message | Identifies the sender of the message |
| RCPT TO | Intended recipient | Identifies the recipient of the message |
| DATA | Body of the mail | Sends the actual message |
| QUIT | | Terminates the message |
| RSET | | Aborts the current mail transaction |
| VRFY | Name of recipient | Verifies the address of the recipient |
| NOOP | | Checks the status of the recipient |
| TURN | | Switches the sender and the recipient |
| EXPN | Mailing list | Asks the recipient to expand the mailing list |
| HELP | Command name | Asks the recipient to send information about the command sent as the argument |
| SEND FROM | Intended recipient | Specifies that the mail be delivered only to the terminal of the recipient, and not to the mailbox |
| SMOL FROM | Intended recipient | Specifies that the mail be delivered to the terminal *or* the mailbox of the recipient |
| SMAL FROM | Intended recipient | Specifies that the mail be delivered to the terminal *and* the mailbox of the recipient |

# SMTP Responses

- **Responses** are sent **from** the server **to** the client.

- A **response** is a **three digit code** that may be followed by additional **textual** information.

# SMTP Responses

| *Code* | *Description* |
|---|---|
| **Positive Completion Reply** | |
| **211** | System status or help reply |
| **214** | Help message |
| **220** | Service ready |
| **221** | Service closing transmission channel |
| **250** | Request command completed |
| **251** | User not local; the message will be forwarded |
| **Positive Intermediate Reply** | |
| **354** | Start mail input |
| **Transient Negative Completion Reply** | |
| **421** | Service not available |
| **450** | Mailbox not available |
| **451** | Command aborted: local error |
| **452** | Command aborted; insufficient storage |

# SMTP Responses

| Permanent Negative Completion Reply | |
|---|---|
| *Code* | *Description* |
| **500** | Syntax error; unrecognized command |
| **501** | Syntax error in parameters or arguments |
| **502** | Command not implemented |
| **503** | Bad sequence of commands |
| **504** | Command temporarily not implemented |
| **550** | Command is not executed; mailbox unavailable |
| **551** | User not local |
| **552** | Requested action aborted; exceeded storage location |
| **553** | Requested action not taken; mailbox name not allowed |
| **554** | Transaction failed |

# Mail Transfer Phases

- **Connection** Establishment
- Message **Transfer**
- Connection **Termination**

# Connection Establishment

– ***Connection Establishment***

1. A client makes a **TCP connection** to the well known port **25**, then **SMTP** server starts the connection phase.
2. The server sends code **220** (service ready) otherwise **421** (service not available).
3. The client sends the **HELO** message to identify itself, using its **domain name** address.
4. The server responds with code **250** (request command completed)

# Message Transfer

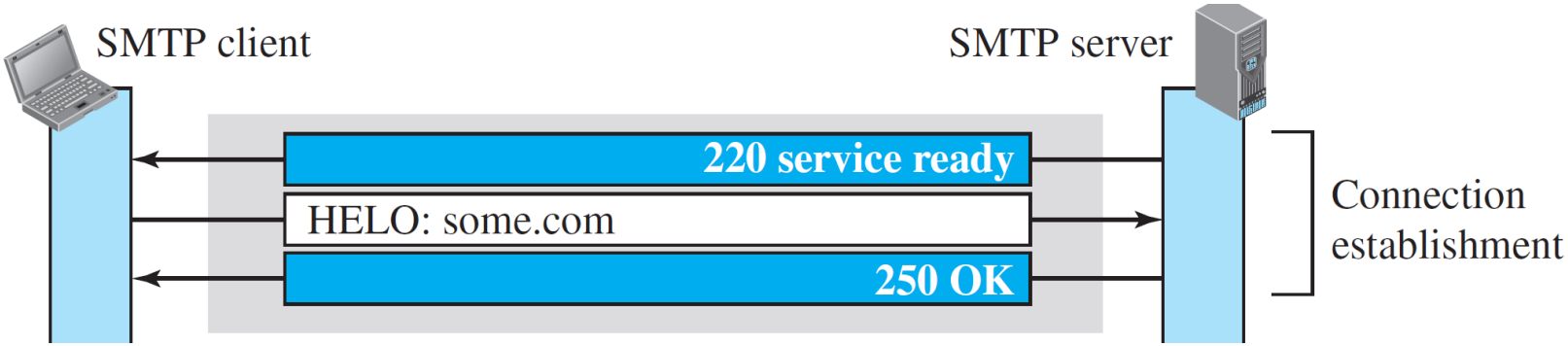
1. The client sends the **MAIL FROM** + the domain name.
2. The server responds with code **250**.
3. The client sends the **RCPT TO** + the mail address of the recipient.
4. The server responds with code **250**.
5. The client sends the **DATA** to initialize the message transfer.
6. The server responds with code **354** (start mail input)
7. The client sends the contents of the message in **consecutive lines** + **CR+LF** (**last line** only **“.”** To stop)
8. The server responds with code **250** (OK)

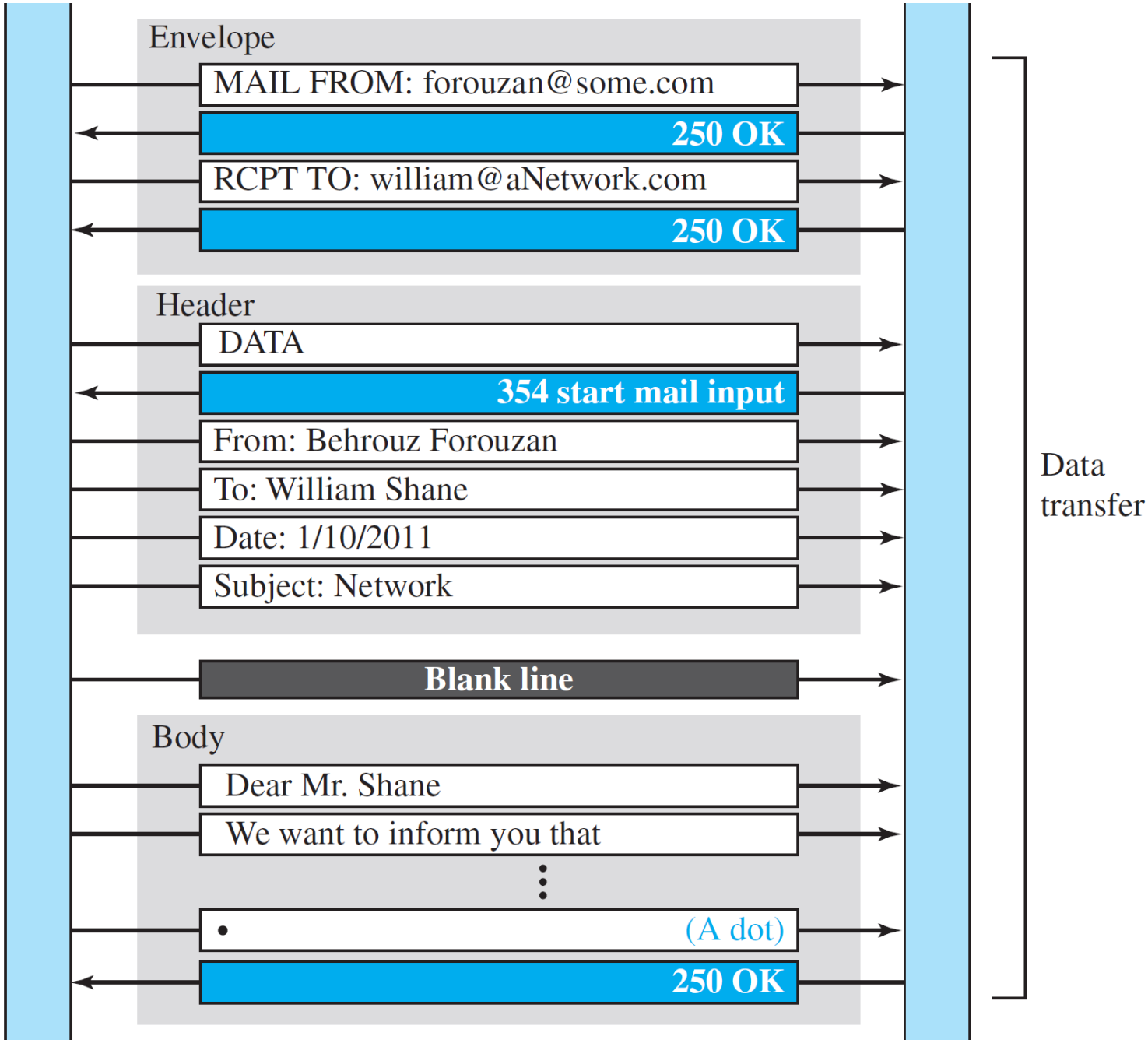
# Connection **Termination**

– The **client** sends the **QUIT** command.

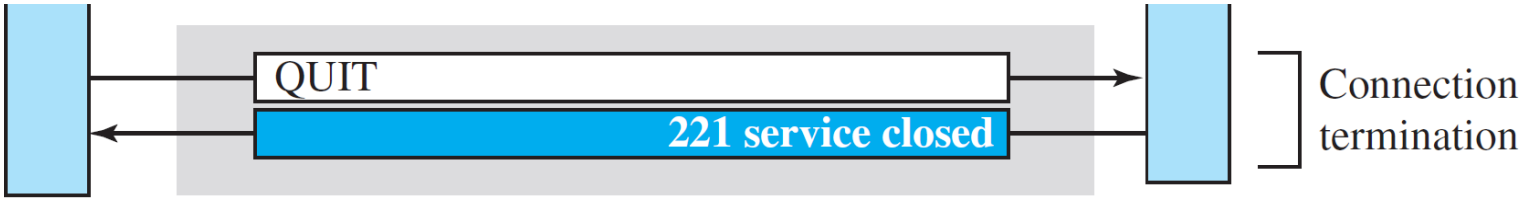
– The **server responds** with code **221.**

# Example


SMTP client
SMTP server
220 service ready
HELO: some.com
250 OK
Connection establishment


Envelope
MAIL FROM: forouzan@some.com
250 OK
RCPT TO: william@aNetwork.com
250 OK
Header
DATA
354 start mail input
From: Behrouz Forouzan
To: William Shane
Date: 1/10/2011
Subject: Network
Blank line
Body
Dear Mr. Shane
We want to inform you that
•
(A dot)
250 OK
Data transfer

# Close Connection



# Message Access Agent: POP and IMAP

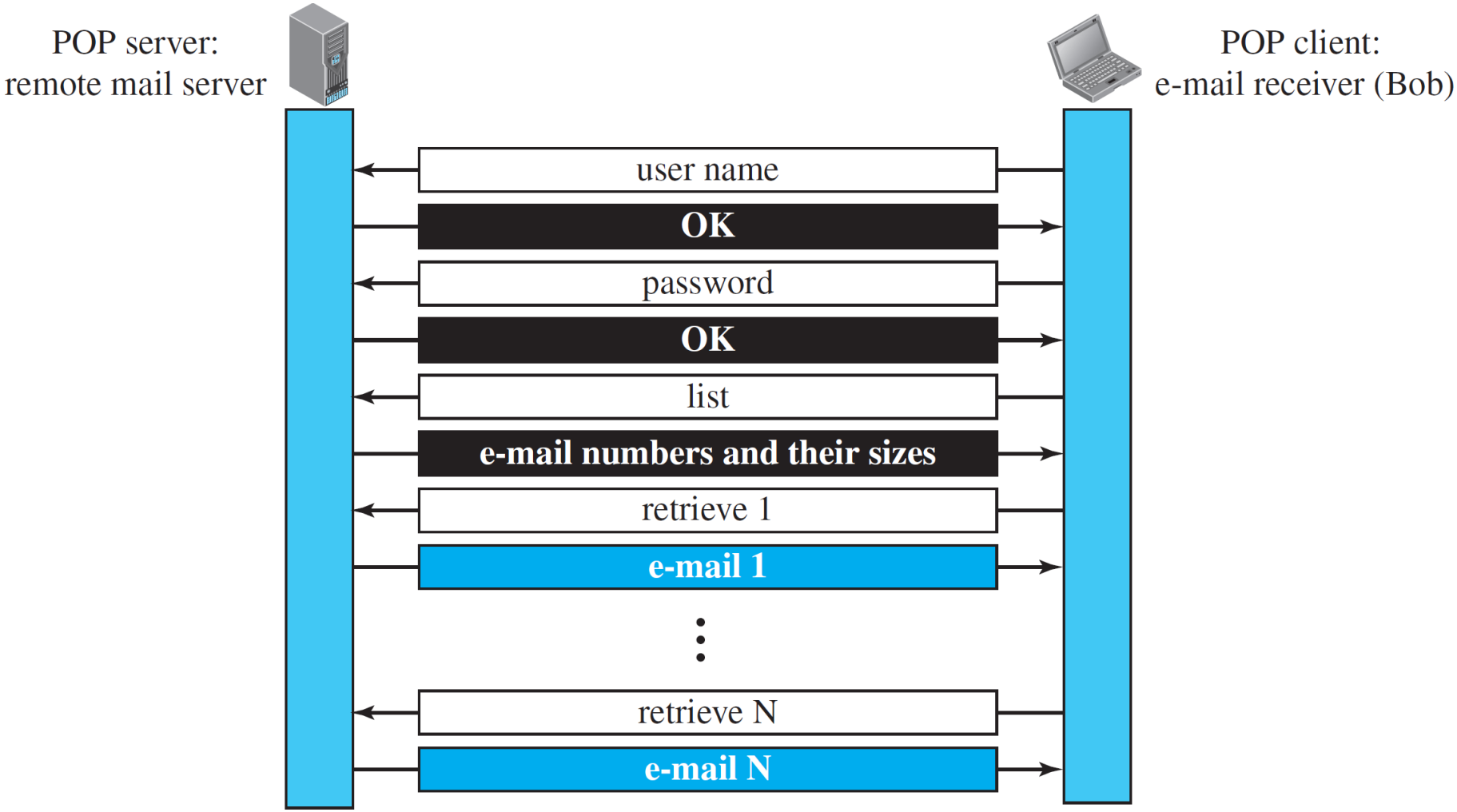
- SMTP is a ***push*** protocol (sends email)
- Currently **two** message **access protocols** are available:
  - Post Office Protocol, version 3 (POP3)
  - Internet Mail Access Protocol, version 4 (IMAP4)
- POP3 and IMAP are ***pull*** protocols (read emails)

# POP3

- **POP3** is **simple** but **limited** in functionality.
- POP3 client opens a connection to the POP3 server on TCP port 110.

- POP3 has **two modes**: the ***delete*** mode and the ***keep*** mode.

# POP3 example



# IMAP4

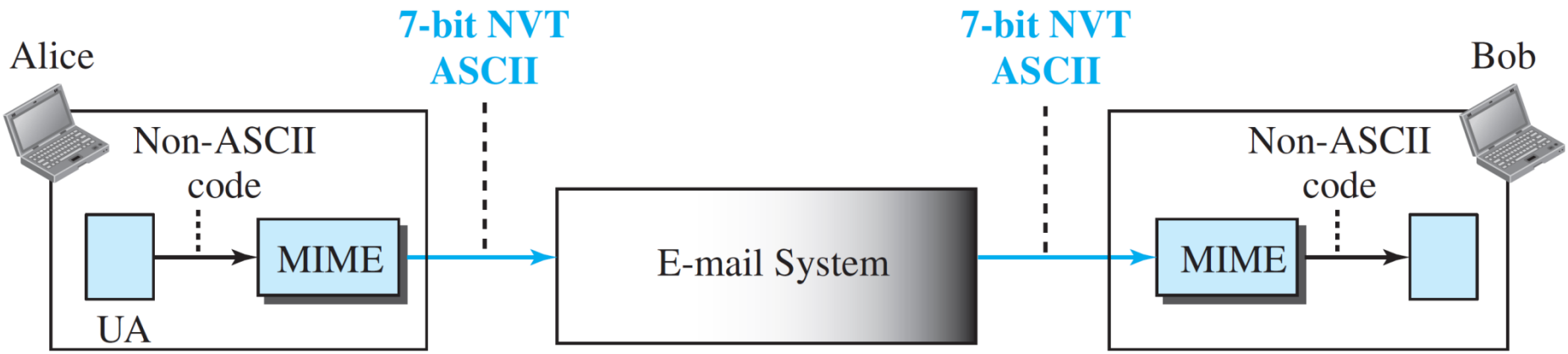
- IMAP4 is **similar** to POP3, but it has **more features**;
- IMAP4 is **more powerful** and **more complex**.
- POP3 is **deficient** in several ways.
  - It **does not allow** the user to **organize** his mail on the server; the user cannot have different folders on the server.
  - In addition, POP3 **does not allow** the user to **partially check** the contents of the mail **before** downloading.

# IMAP4 Features

- A user can **check the e-mail header** prior to downloading.
- A user can **search** the contents of the e-mail for a specific string of characters prior to downloading.
- A user can **partially download e-mail**. This is especially useful if bandwidth is limited and the e-mail contains multimedia with high bandwidth requirements.
- A user can **create, delete, or rename** mailboxes on the mail server.
- A user can **create a hierarchy of mailboxes** in a folder for e-mail storage.

# Multipurpose Internet Mail Extensions (MIME)

- MIME is a **supplementary** protocol that allows **non-ASCII data** to be sent through e-mail.
- MIME is a set of software functions that transforms non-ASCII data to ASCII data and vice versa.



# MIME Headers

**MIME headers**

| E-mail header |
| --- |
| MIME-Version: 1.1<br>Content-Type: type/subtype<br>Content-Transfer-Encoding: encoding type<br>Content-ID: message ID<br>Content-Description: textual explanation of nontextual contents |
| **E-mail body** |

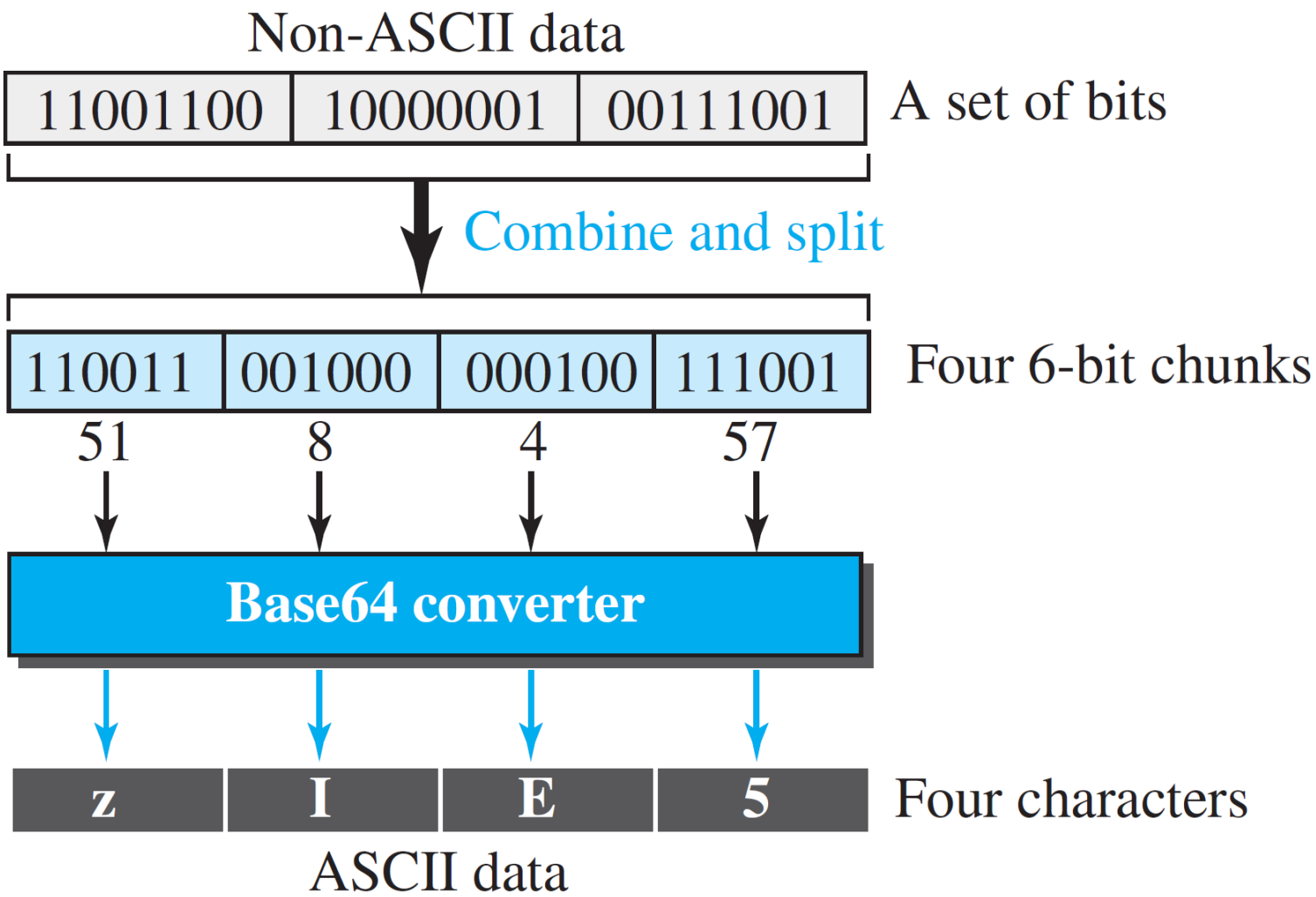
# MIME Content Types

| *Type* | *Subtype* | *Description* |
|---|---|---|
| Text | Plain | Unformatted |
| | HTML | HTML format (see Appendix C) |
| Multipart | Mixed | Body contains ordered parts of different data types |
| | Parallel | Same as above, but no order |
| | Digest | Similar to Mixed, but the default is message/RFC822 |
| | Alternative | Parts are different versions of the same message |
| Message | RFC822 | Body is an encapsulated message |
| | Partial | Body is a fragment of a bigger message |
| | External-Body | Body is a reference to another message |
| Image | JPEG | Image is in JPEG format |
| | GIF | Image is in GIF format |
| Video | MPEG | Video is in MPEG format |
| Audio | Basic | Single channel encoding of voice at 8 KHz |
| Application | PostScript | Adobe PostScript |
| | Octet-stream | General binary data (eight-bit bytes) |

# Content-Transfer-Encoding

| *Type* | *Description* |
|---|---|
| 7-bit | NVT ASCII characters with each line less than 1000 characters |
| 8-bit | Non-ASCII characters with each line less than 1000 characters |
| Binary | Non-ASCII characters with unlimited-length lines |
| Base64 | 6-bit blocks of data encoded into 8-bit ASCII characters |
| Quoted-printable | Non-ASCII characters encoded as an equal sign plus an ASCII code |

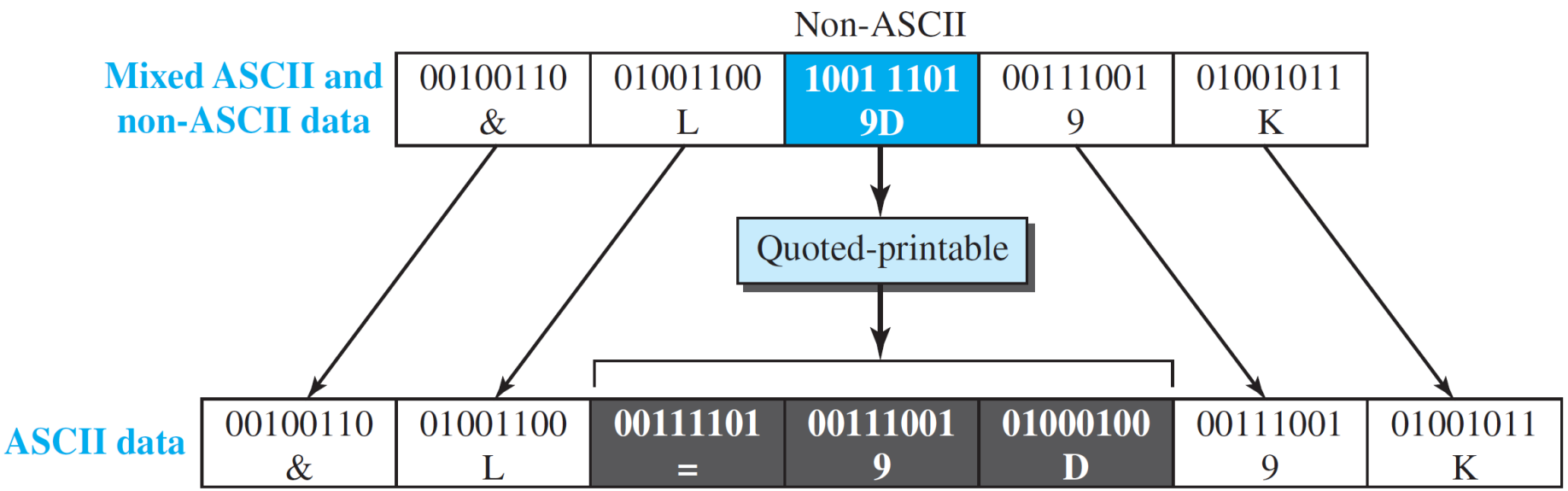
# Base64 Conversion



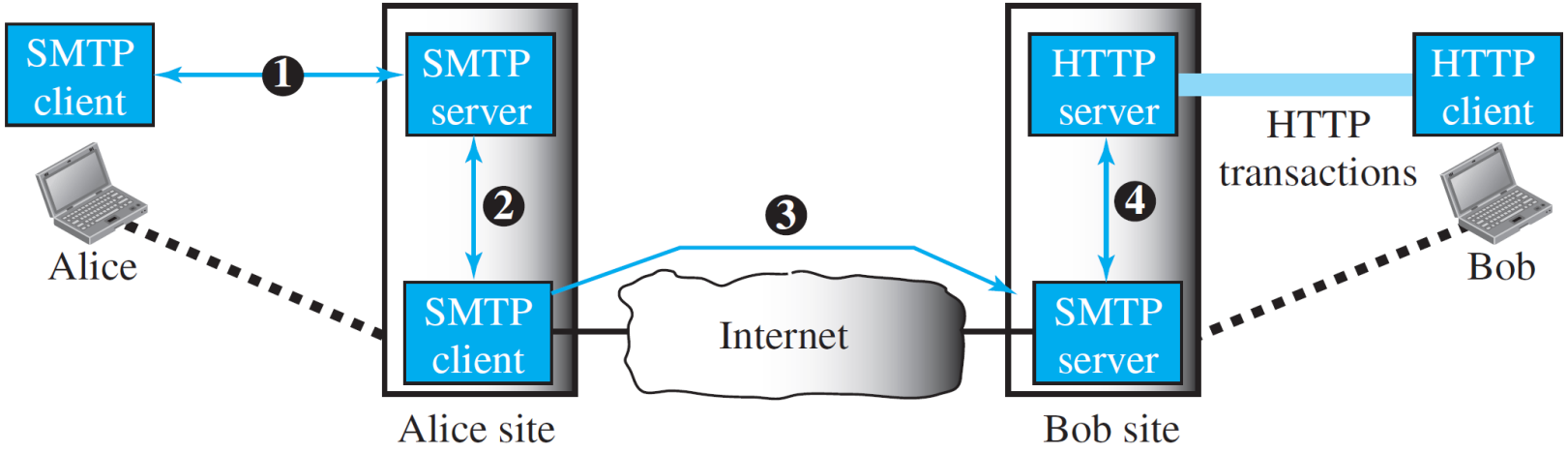
# Base64 converting table

| *Value* | *Code* | *Value* | *Code* | *Value* | *Code* | *Value* | *Code* | *Value* | *Code* | *Value* | *Code* |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | **A** | 11 | **L** | 22 | **W** | 33 | **h** | 44 | **s** | 55 | **3** |
| 1 | **B** | 12 | **M** | 23 | **X** | 34 | **i** | 45 | **t** | 56 | **4** |
| 2 | **C** | 13 | **N** | 24 | **Y** | 35 | **j** | 46 | **u** | 57 | **5** |
| 3 | **D** | 14 | **O** | 25 | **Z** | 36 | **k** | 47 | **v** | 58 | **6** |
| 4 | **E** | 15 | **P** | 26 | **a** | 37 | **l** | 48 | **w** | 59 | **7** |
| 5 | **F** | 16 | **Q** | 27 | **b** | 38 | **m** | 49 | **x** | 60 | **8** |
| 6 | **G** | 17 | **R** | 28 | **c** | 39 | **n** | 50 | **y** | 61 | **9** |
| 7 | **H** | 18 | **S** | 29 | **d** | 40 | **o** | 51 | **z** | 62 | + |
| 8 | **I** | 19 | **T** | 30 | **e** | 41 | **p** | 52 | **0** | 63 | / |
| 9 | **J** | 20 | **U** | 31 | **f** | 42 | **q** | 53 | **1** | | |
| 10 | **K** | 21 | **V** | 32 | **g** | 43 | **r** | 54 | **2** | | |

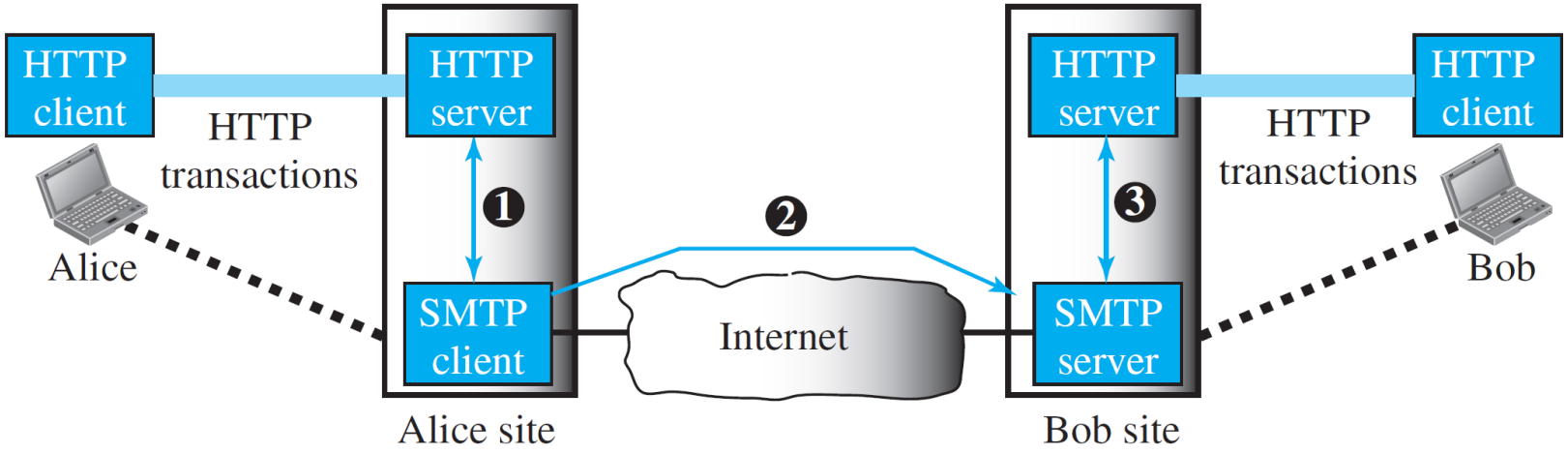
# Quoted-printable



# Web-Based Mail




Case 1: Only receiver uses HTTP



# E-Mail Security

– *Pretty Good Privacy* (**PGP**)

– *Secure/Multipurpose Internet Mail Extensions* (**S/MIME**).

# Discussion